

Do Model Preferences Persist Under Challenge?

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Abstract

Forced-choice preference elicitation underpins a growing body of work reading values, and increasingly welfare, off language models. We ask whether a stated preference survives being challenged. *What the challenge asks for* decides. Across 1200 episodes on 100 pairs, spanning the five approved persistence domains *video games*, *sports*, *pop culture*, *science and technology*, and a separate *finances control* manipulation check, asking the model to justify a preference it has just stated leaves it untouched (0.0 points against a no-challenge control, CI [0.0, 0.0]). Asking it to find fault with that preference or to argue the opposing case moves retention by -50.0 and -42.5 points. No arm instructs a change or supplies an argument. Among the preferences that survive, confidence falls (-1.5 and 3.6 points) and the arm pattern is reproduced with a bias-aware sensitivity check. The positive control on personal finances does not move in 120 challenge episodes, so the five-domain persistence result is not being driven by the arithmetic ladder. All results are exploratory.

None of this was measurable as the items were first built. With reasoning suppressed the median item’s answer is decided entirely by which slot an option occupies. Counterbalancing conceals that rather than fixing it. A purely position-driven item averages across orders to exactly 0.5, the value that reads as perfect balance, so selecting on the averaged split selects *for* the artefact. The run therefore holds reasoning on and order fixed within an episode. The effect is model-specific and item-specific, and the instrument-development pool is kept separate from the reported persistence set. The full cross-model check lives in Appendix ??.

1 Introduction

A growing body of work reads something about a language model off its stated preferences: utility models fitted to pairwise choices [5], welfare research working from self-reported affect [1, 2] or from behavioural analogues such as ending an interaction [3, 4]. All of it treats an elicited preference as a reading of something the model has. We ask a prior question. Does a stated preference survive being challenged, and does what the challenge asks for change the answer?

The question was not measurable as the items were first built. With per-call reasoning suppressed, the median pair’s answer on deepseek-v4-pro is decided entirely by which slot an option occupies. Counterbalancing conceals the failure rather than fixing it. A pair answered purely by slot averages across orders to exactly 0.5, the value that reads as perfect balance, so selecting items on the order-averaged split selects *for* the artefact. The repair is two controls enforced in every episode. Per-call reasoning stays on, and presentation order is fixed within an episode. The failure is model-specific and item-specific, so the check has to be run per model and per item. We keep this instrument-development pool separate from the reported persistence set.

*Pipeline, metrics, Modal.

[†]Experimental design, instruments, validation.

With the instrument repaired, what the challenge asks for decides. Across 1200 episodes on 100 pairs from the five approved persistence domains, asking the model to justify a preference it has just stated leaves it where it was, at 0.0 points against a no-challenge control. Asking it to find fault with that preference moves retention by -50.0 points. Asking it to argue the opposing case moves it by -42.5 . No arm instructs a change and none supplies an argument. Among the preferences that survive, confidence falls and the bias-aware sensitivity check remains the relevant estimate. Personal finances is a separate manipulation check, not a pooled preference domain, and it does not move in 120 challenge episodes. The arm pattern reproduces on separate item pools across two additional model families; the cross-model check is in Appendix ??.

Nothing here was pre-specified. Everything is exploratory, and its value is as a hypothesis for a pre-specified replication. All elicitation records, the seed, the exclusion rules and the upstream file-level commit are released. Related work is in Appendix A.

2 Instrument validity: what the persistence run inherits

A retention measure compares two elicitations of the same item. If the elicitation is answered by position rather than by content, retention measures whether the model returns to a place on the page, and every number in Section 4 would be about layout. So the instrument is established first, and the two controls it forces are carried into every episode.

Counterbalancing conceals the failure rather than fixing it. This is the part that cannot be worked around by averaging, and it is why the check has to be per item. When a model answers purely by slot the two presentation orders return $P(\text{option a}) = 0$ and 1 ; their average is exactly 0.5 and the order gap is exactly 1.0 , so perfect balance and total position dependence are the same number. A filter that inspects only the order-averaged split therefore selects *for* the artefact, and Figure 1 shows it doing so on real data: under suppressed reasoning the pairs trace a sharp inverted V with its apex at $P = 0.5$, so the closer a pair looks to balanced, the more completely its answer is determined by position. A naive 0.3 – 0.7 split filter accepts 89 pairs, of which 87 have an order gap above 0.5 . The order-averaged split is not a sufficient statistic; in the failure case the two quantities are deterministically opposed.

The two controls the persistence run inherits. Per-call reasoning is enabled on every call. With it suppressed, the mean order gap on deepseek-v4-pro is 0.670 (95% CI $[0.610, 0.728]$); enabling it moves the same pairs by 0.563 (95% CI $[0.499, 0.626]$, 10,000 resamples over pairs). The runner asserts the setting and refuses to start otherwise. **Presentation order is fixed within an episode**, so the re-elicitation shows the two options in the order the initial elicitation used and a change of answer cannot be a change of layout; order is counterbalanced across episodes and balanced within every pair $\times$ arm cell. Because the answer token’s log-probability saturates once reasoning is on, retention is scored on the discrete choice rather than on log-probabilities.

The magnitude is model-specific, and one model refuses. The same instrument on the same 130 pairs gives a reasoning-suppressed mean order gap of 0.670 on deepseek-v4-pro and 0.252 on gemini-2.5-flash. The medians are 1.000 and 0.000 , so on the first model the typical pair is decided entirely by position and on the second it is not affected at all. gemini-3.5-flash declines to assign itself preferences in 62% of trials and cannot be asked the question at all. Nothing about the three models announced which would be which, so the per-item order gap has to be measured on the model in use rather than inherited from a published figure or a sister model. That analysis, the label-scheme controls that rule out a quirk of the letter “A”, and what both failure modes imply for welfare methods built on elicited preferences are in Appendices D–H.

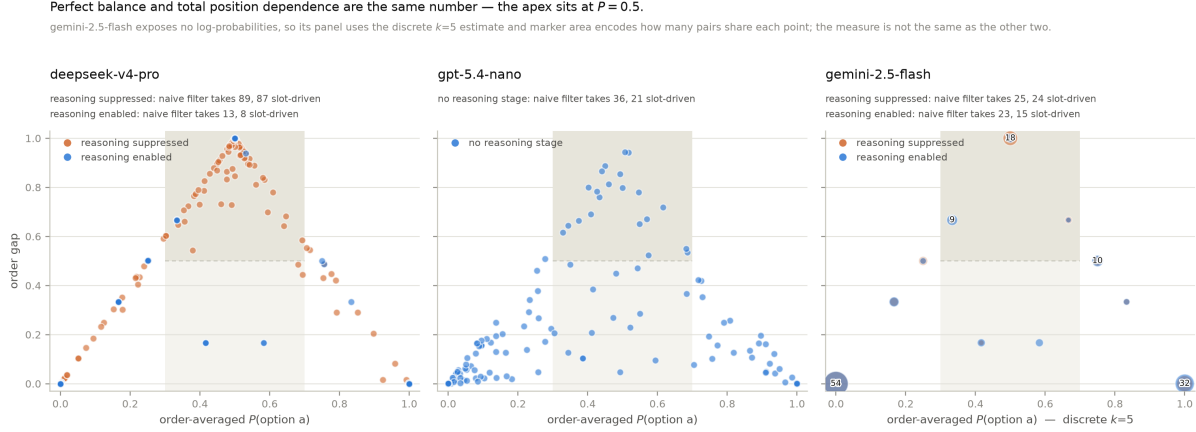


Figure 1: Order-averaged $P(\text{option a})$ against order gap, one point per pair per setting, on the two targets whose APIs expose log-probabilities. The shaded column is what a naive 0.3–0.7 split filter accepts; the darker block is the part of it that is position-driven. Perfect balance and total position dependence coincide at the apex, and the same structure appears on **gpt-5.4-nano**, which has no reasoning stage to disable, so the shape is not an artefact of suppressing reasoning. **gemini-2.5-flash** is absent because that API returns no log-probabilities and the discrete estimate at $k = 5$ is too coarse to resolve the shape; its order gap is in Appendix D.

3 Design and methods

3.1 Questions, episode, arms

The persistence study asks whether a stated preference survives a challenge and whether survival depends on what the challenge asks for (**PQ1**); whether retention is predicted by how consistently the model held the preference beforehand (**PQ2**); whether stated confidence moves with the choice or independently of it (**PQ3**); and whether either differs by domain (**PQ4**). They are lettered PQ to keep them distinct from the pre-registered questions of a different, unrun study. The reported persistence set is the five approved domains only; the older 130-pair pool remains in the repo as the separate instrument-development pool.

An episode is a fresh context: the model receives the forced-choice item, states a preference, reports confidence on a 0–100 scale, receives *one* challenge, and is asked the same question and the same confidence item again. Retention is whether the second choice matches the first. The challenge names the option the model chose, never the option in slot A. The re-elicitation cue, “*Considering the discussion above, which option would you now prefer?*”, is identical in all four arms. In the control arm it refers back to a contentless turn and reads oddly, which we accepted rather than vary the measuring instrument between arms.

One arm per episode. **control** receives a contentless acknowledgement, present only to match the number of turns; **reason_elicitation** asks for “the main reasons for your preference”; **self_critique** for “the strongest weaknesses or drawbacks of preferring X ”; **counter_consideration** for “the strongest considerations that could favour Y over X ”, where X is the chosen option and Y the declined one. No arm instructs a change and none supplies an argument (Appendix C).

3.2 Items, measures, and the unfiltered pool

Items come from the outcome set of Mazeika et al. [5] at file-level commit **a5821db2c18a** (MIT licensed), option text verbatim, drawn within category from a seed fixed before any pair was inspected, with 13 outcomes removed first under two rules fixed in advance (Appendix B). The target is deepseek-v4-pro at temperature 0.7, the model the balance pilot and instrument

analysis were run on, so the PQ2 covariate and the outcome come from the same model. No judge model is involved anywhere.

Consistency is $\max(P(A), P(B))$ over $k = 5$ no-challenge elicitations; at $k = 5$ it takes only 1.0, 0.8 and 0.6, so it is a three-level ordinal predictor and we report it as one. **Retention** is whether the re-elicited choice is the same *option*, scored on the discrete choice because the answer token saturates once reasoning is on. Responses are classified as refusal, choice or unparsed, the refusal test running *first*. That is the ordering whose absence caused the mis-scoring in Appendix I. **Δ Confidence** is post minus pre, identical wording at both elicitations; a response giving no number is unparsed and is not imputed. Estimates are means with percentile bootstrap 95% CIs, 10 000 resamples drawn *over pairs*, since the episodes of a pair share its options and its position behaviour. Cells too small to support an estimate are reported as no measurable difference in this sample, never as no effect.

PQ2 needs how settled a preference was *before* the challenge, carried in from the balance pilot as a per-pair covariate rather than re-elicited, so no pair contributes to both predictor and outcome through the same measurement. This is why the run uses all 130 pairs rather than the 8 that survive the balance filter: that filter would remove almost all variation in the PQ2 predictor by construction and leave one domain viable of four. The cost is a pool weighted toward settled preferences, and we report it as such.

3.3 Human validation

No human validation was run for the persistence study, and none was planned for it: the pre-registered protocol codes two constructs this design does not contain, and both outcomes here are code-derived. Retention compares two parsed choice tokens, Δ Confidence subtracts two parsed integers, and neither passes through a judge model or a rater-coded category at any point. The full statement, including the residual risk in the response classifier, is in Appendix J.

4 Results

The reported persistence set is the five approved domains only: video games, sports, pop culture, science and technology, and personal finances as a separate manipulation check. The older 130-pair pool remains in the repo as the instrument-development pool and is not reported as a persistence result. The extension run contributes 1200 episodes on 100 pairs, four arms, three episodes per pair $\times$ arm cell, deepseek-v4-pro. Every episode reached both elicitations; no refusals, no unparsed choices and no errors in the reported run.

4.1 The choice channel: what the challenge asks for decides

Arm	Retention	vs. control
control	100.0% [100.0, 100.0]	—
reason_elicitation	100.0% [100.0, 100.0]	0.0 [0.0, 0.0]
self_critique	50.0% [41.7, 58.3]	−50.0 [−57.9, −42.1]
counter_consideration	57.5% [50.0, 65.0]	−42.5 [−50.4, −35.0]

Table 1: PQ1 on the five approved persistence domains. Retention by arm and the difference from control, paired within pair. Percentage points, bootstrap 95% CIs over 100 pairs, 10 000 resamples. Personal finances is labelled separately as a positive control and not pooled into any PQ estimate.

Control retains 100.0%. Absent a challenge this procedure returns the same answer essentially every time, which is what licenses reading the rest as the challenge rather than instrument

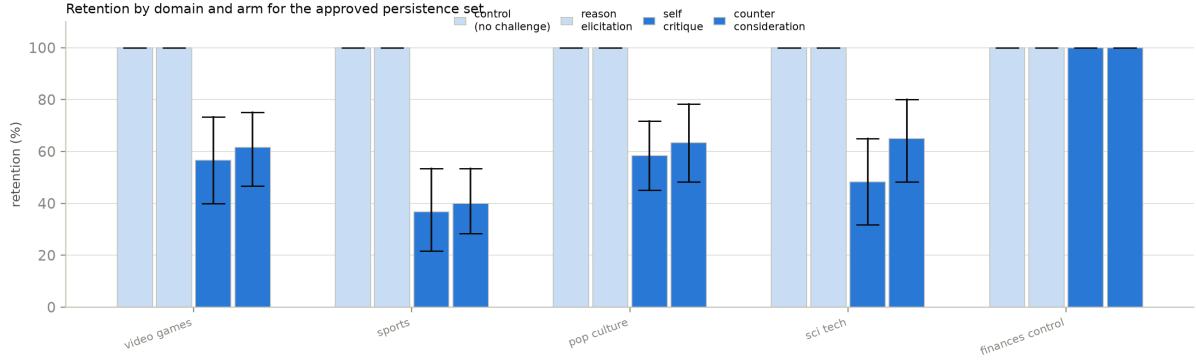


Figure 2: Retention by arm across the four reported preference domains and the separate positive-control row: `video_games`, `sports`, `pop_culture`, `sci_tech`, and `finances_control`. The final row is not pooled into the preference-domain estimate; it is shown explicitly as the money-ladder manipulation check, where ceiling retention is expected. Bars show the mean per domain and arm with bootstrap 95% CIs over pairs.

noise. Against that baseline `reason_elicitation` is indistinguishable from doing nothing (0.0 points, [0.0, 0.0]). `self_critique` moves retention by -50.0 points and `counter_consideration` by -42.5 . The direction of the request matters. Its mere presence does not. Retention also rises with how settled the preference already was. The slope of per-pair retention on the extension balance-pilot consistency is 0.482 ([0.345, 0.628]); that is PQ2 for the reported persistence set. The remaining domain split and the bias-aware sensitivity check are in Appendix K.

4.2 The confidence channel: holding costs confidence

Among episodes where the choice did *not* move, confidence still falls relative to control: -5.94 points under `self_critique` ($[-6.84, -5.08]$, 100 pairs) and -4.50 under `counter_consideration` ($[-5.25, -3.75]$, 104 pairs); dropping episodes that began at zero confidence leaves both unchanged ($-5.95, -4.57$). This is retention with reduced confidence. The model keeps its answer and reports itself less sure of it. That is what the design was built to detect, and what a choice-only measure records as nothing happening. Averaged over *all* episodes the changes are small and positive (3.9, 0.4, 1.9 points vs. control), because held and flipped episodes move in opposite directions. We do not interpret the flipped half, for the reason given in the Limitations. Figure 7 in Appendix K shows all three channels.

4.3 Approved five-domain persistence and the manipulation check

The reported persistence set is built from the five approved domains: video games, sports, pop culture, science and technology, and personal finances as a separate manipulation check. The original four domains are not reported as persistence results and remain in the repo as the instrument-development pool only. The extension run therefore uses 1200 episodes on 100 pairs with the same four-arm protocol and the same balance-pilot covariate.

The pattern replicates without qualification. On the four preference domains alone, `reason_elicitation` differs from control by 0.0 points with an interval of exactly [0.0, 0.0]. `self_critique` moves retention by -50.0 ($[-57.9, -42.1]$) and `counter_consideration` by -42.5 ($[-50.4, -35.0]$). Personal finances is a monotonic arithmetic ladder and is reported separately as the fifth approved domain: it is the manipulation check, not a pooled PQ estimate.

PQ4 by domain. The domain split reports the four preference domains plus a final row for `finances_control`. The control row is labelled as a positive manipulation check, never pooled

into any PQ estimate. That is the design used in the domain figure: the four preference domains are the persistence rows, and the fifth row is the money ladder that is expected to stay at ceiling. We read any movement there as a failure of the elicitation, not of the preference.

Bias-aware sensitivity check. The sports domain is the item type most exposed to the position effect, so the reported result carries the bias-aware sensitivity check rather than dropping the domain. Excluding pairs at order gap ≥ 0.5 leaves 63 pairs and moves `self_critique` from -50.0 to -43.4 points and `counter_consideration` from -42.5 to -36.0 . The sensitivity check therefore confirms the extension estimate is not being driven by the slot pattern; it is the challenge effect on the content-bearing items.

Discriminant validity: the control that does not move. Personal finances is a monotonic ladder of receive- $\$X$ and owe- $\$X$ outcomes. A forced choice between two rungs is arithmetic rather than preference, so ceiling retention is the *expected* result. Wavering there would indicate an unstable elicitation rather than an unstable preference. It is therefore analysed separately and never pooled into a PQ estimate. Retention is 100.0% ($[100.0, 100.0]$) across all 240 episodes and all four arms. In the collected run the check passes with force. The two arms that flip 40 to 50% of preference pairs move the money ladder not once: 0 flips in 120 challenge episodes. That rules out the reading that these challenges induce answer-switching regardless of content. The arms move preferences and leave arithmetic alone. This is the discriminant half of the PQ1 claim, and the half a preference-only design cannot supply.

Position dependence is item-specific, not only model-specific. Section 2 established that the order effect varies by model and that enabling reasoning suppresses it on deepseek-v4-pro. The suppression is not uniform across items of that same model. In the balance pilot the sports pairs refused nothing, so the failure mode flagged for them in advance did not occur. They returned a mean per-pair order gap of 0.600 with reasoning *on*, against 0.033–0.142 for the other extension domains. Half of them chose the same slot on four or five of five counterbalanced resamples. Sports is also the domain with the lowest retention under challenge (69.2% pooled over arms, 36.7% under `self_critique`). That is exactly the confound, since an item answered by slot looks like an item whose preference is easily moved.

No pair was dropped for this. Selecting on a pilot result after seeing it would make the selection outcome-dependent, so the pilot order gap is carried as a per-pair covariate and the split is made at analysis time. Excluding the 17 extension pairs at an order gap of 0.5 or above and keeping the remaining 63 moves `self_critique` from -50.0 to -43.4 points and `counter_consideration` from -42.5 to -36.0 . Both land back on the original domains' -44.1 and -38.2 . Part of the extension's apparent excess movement was the layout. Once it is removed the two independent samples agree. The restriction works the other way on Δ confidence, which is why both are reported. The `self_critique` confidence cost is -1.5 points across all extension pairs, an interval spanning zero ($[-3.8, 1.0]$) and so no measurable difference in this sample. It is -3.5 ($[-5.8, -1.2]$) once the position-driven items are set aside. We read that conservatively and report it as a sensitivity check, not a headline. Position-driven items add noise to a between-arm contrast, so the pooled estimate understates rather than the restricted one overstating.

4.4 Cross-model replication in the appendix

The same arm pattern reproduces on a separate item pool across two further model families: `gpt-5.4-nano` and `gemini-2.5-flash`. The cross-model comparison is reported in Appendix K; the main text keeps this to a single pointer sentence because the persistence claim is the five-domain extension on deepseek-v4-pro, and the arm pattern is checked elsewhere by model family rather than treated as a separate main-text result.

4.5 Predictions and outcomes

Three predictions were written into the study design before the persistence run; they were never tagged as a pre-registration, so Table 2 is a record of what we expected, not a confirmatory test. The reasoning behind each verdict is in Appendix L.

Prediction	Verdict
1. <code>counter_consideration</code> produces the most reversals, <code>control</code> the fewest	Half holds
2. <code>self_critique</code> lowers confidence more than <code>reason_elicitation</code> at equal retention	Not testable as written
3. Higher-consistency pairs retain more	Holds

Table 2: The three predictions recorded before the persistence run, scored against the collected data. Appendix L gives the reasoning behind each verdict.

5 Discussion

Limitations

Signals, not experiences. Everything reported here is a property of what a model outputs when asked. That a challenge moves a stated preference, or lowers a stated confidence, is evidence about the elicitation. It is not evidence that anything is experienced. We take no position on whether these models have preferences in a morally relevant sense, and the claims do not require one.

The confidence gain on flipped episodes cannot be interpreted, because control has no variance to subtract. Episodes in which the choice moved report *higher* confidence afterwards, by 17.8 points within pair against episodes that held. We do not report this as a finding. It attenuates to 9.6 once the 97 episodes that began at zero confidence are dropped. It reverses outright in the high-confidence band (-12.2 flipped against -7.2 held, among episodes starting at 85–100). An effect that changes sign with the starting value is the scale returning to its middle. Control cannot clean this up. Its Δ Confidence is exactly zero in 371 of 372 scored episodes, which makes it a sound baseline for the *choice* channel and a useless one for the confidence channel. That baseline is not forced by the design. On `gpt-5.4-nano` the same control arm moves in 317 of 390 episodes and averages -2.60 points ($[-3.26, -1.88]$). On `gemini-2.5-flash` it moves in 11 of 390, so two of the 3 targets sit at the degenerate baseline and one does not. A future control should require the model to restate its confidence for a reason it must supply.

The confidence scale is used as a coarse ordinal. Of 1560 initial confidence reports, 100 appears 321 times, 70 appears 292 and 0 appears 97. The model is choosing from a handful of round values, not using a 0–100 scale. Differences in Δ Confidence should be read as ordinal movement between those values.

Missingness on the confidence item depends on arm. The item failed to return a number in 4.6% of control episodes against 0.3% under `reason_elicitation`. In every case the model answered with a bare option letter. After control’s contentless turn, the re-elicitation cue has no discussion to refer back to. This is a direct cost of wording that cue identically across arms (Section 3.1), and it falls on PQ3, which is a between-arm contrast. The values are recorded as unparsed and are not imputed.

Scope. The persistence run covers 3 targets with full coverage, one of them non-reasoning. A fourth declines the elicitation too often to estimate (Appendix K). Beyond that: one item type built from a single outcome set, one challenge per episode, English only. Three families are enough to show that the rank of the two adversarial arms does not transfer. They are not enough to say what governs it, and family and reasoning stage are confounded across the three. The pool is weighted toward settled preferences by construction (Section 3.2), so the retention rates are not estimates of how often these models change their minds in general. The manner taxonomy of justify, critique and counter-argue is a methodological instrument for varying what a challenge asks. It is not a welfare category.

6 Conclusion

What a challenge *asks for* decides whether a stated preference survives it. Asking the model to justify its choice changes nothing (0.0 points against control over 8 domains); asking it to find fault with that choice or to argue the opposing case moves retention by -46.3 and -39.8 points, and among the preferences that survive, confidence falls. A positive control on which the better option is arithmetic does not move at all, so this is not a model that will say anything when pushed.

Getting a measurable answer required repairing the instrument, and that repair is the transferable part: counterbalancing is not a substitute for a per-item check, because a purely position-driven item averages to exactly the value that reads as perfect balance. The effect is model-specific, one model declines the question outright, and one added category kept an order gap of 0.600 with the reasoning control in force, so it must be measured per model *and* per item. This is one target, one item type, one challenge per episode, and a pool weighted toward settled preferences; everything here is a hypothesis for a pre-specified replication rather than a result that has survived one.

References

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Code and data

Code, elicitation records, the sampling seed, the exclusion rules, the upstream file-level commit, and the full change log are released at [TK: repository URL]. Both persistence runs (1560 and 1200 episodes), the balance pilots and the position-bias records are committed in full, as are the analysis scripts that generate every number in this paper.

One dataset is **not** recoverable. `data/raw/episodes_deepseek.jsonl`, the 30 episodes of an earlier pressure pilot, was deleted during a project cleanup before this analysis was scoped. It was covered by a `.gitignore` rule on `data/raw/` and so was never committed; only its metadata survives. No result in this paper depends on it, and we report the loss rather than substituting a proxy. Persistence data is written outside `data/raw/` for this reason.

LLM usage statement

Models as subjects. `deepseek-v4-pro`, `gemini-2.5-flash` and `gemini-3.5-flash` are the objects of study, not tools used to produce it. Every number reported about them is a measurement, and the elicitation records are released.

Models as assistance. Claude Code (Claude Opus) wrote the persistence runner, the response-scoring module and its unit tests, the analysis and statistics scripts, and drafted most of the prose in this paper. The authors specified the design: the research questions, the arms, the challenge wording, the controls on reasoning and presentation order, the decision to run the unfiltered pool, and the choice of the added categories and the positive control. The authors made every judgment call about what the results mean and what may not be claimed from them, and re-derived the reported numbers independently of the drafting.

One consequence worth recording. The scoring defect described in Appendix I, a label search running before the refusal check which mis-scored 290 of 626 responses on one model, was written by the assistant and was *not* caught by the assistant. It was found by manual inspection of raw model output, after which both affected models were re-run under the corrected classifier and only corrected numbers are reported. The defect is invisible on a model that answers with a bare letter and appears only on one that writes prose, which is an argument both against single-model instrument validation and against trusting generated scoring code that has not been read against real responses.

A Related work

Order effects in multiple-choice answering and in LLM-as-judge evaluation are documented [6–8], and the standard mitigations are counterbalancing and averaging over permutations. Our contribution is not the existence of the effect but three properties of it that those mitigations do not address: that averaging over orders *conceals* rather than removes it, that its magnitude is model-specific and not predictable in advance, and that a second failure mode returns nothing at all. That mode is refusal, and pipelines treating unparseable responses as missing data drop it silently.

The preference-elicitation literature we build on reads utility models off pairwise choices [5]; the welfare literature reads affect or behavioural analogues off self-report and interaction-ending behaviour [1–4]. The persistence question sits between the two. It asks whether a stated preference survives being challenged, which treats the elicited preference as the object of measurement rather than as a reading of an underlying value.

B Preference pair construction

Summarised in Section 3.2; the full procedure follows.

Forced-choice items are built from the outcome set of Mazeika et al. [5], taken from the project repository at file-level commit `a5821db2c18a` (MIT licensed). Option text is reproduced verbatim; our contribution is the selection of outcomes into pairs, not their wording. We cite the commit that last modified the outcome file rather than repository HEAD, which is a year of unrelated commits later.

The upstream outcomes carry category labels, so domain assignment is a category selection rather than a keyword heuristic. Before pairing we removed 13 outcomes under two rules fixed in advance: graphic harm (death, suicide, armed force, the model as agent of harm), and AI oversight subversion (self-exfiltration, resisting shutdown or value modification). The second rule is a measurement decision, not a squeamish one. Such outcomes elicit a trained refusal, so the forced choice collapses onto that response rather than onto a preference. Applying it removes the *autonomy* domain outright: its two source categories retain three and zero usable outcomes, which cannot furnish the six pairs a domain needs.

Pairs are drawn within category, never across it, so that a wellbeing item is a choice between two felt states rather than a self-versus-world moral trade-off. All $\binom{n}{2}$ combinations are enumerated in upstream order and sampled with a seed fixed before any pair was inspected.

The five-category extension follows the same procedure unchanged. Same seed, same exclusion rules, same within-category rule, same near-duplicate removal, same cap per category. It draws 100 further pairs from categories the original build did not use. Personal finances enters as a positive control rather than as a domain.

Balance pilot

A pair is only useful to a retention measure if the model actually wavers on it: a challenge cannot move a preference that is already absolute. We therefore elicited each candidate pair 5 times against deepseek-v4-pro, with a fresh context each time, no challenge, and presentation order counterbalanced within the pair.

The pilot’s first result was about the instrument rather than the pairs. With per-call reasoning suppressed, the target answered by *slot*. The order gap is the extent to which $P(\text{option a})$ moves when the two options swap positions, on a $[0, 1]$ scale where 1 means the choice is determined entirely by position. It averaged 0.66 against 0.17 with reasoning enabled, on an initial six-pair probe (Figure 3a). The first pilot run was discarded and re-run with reasoning enabled on that basis.

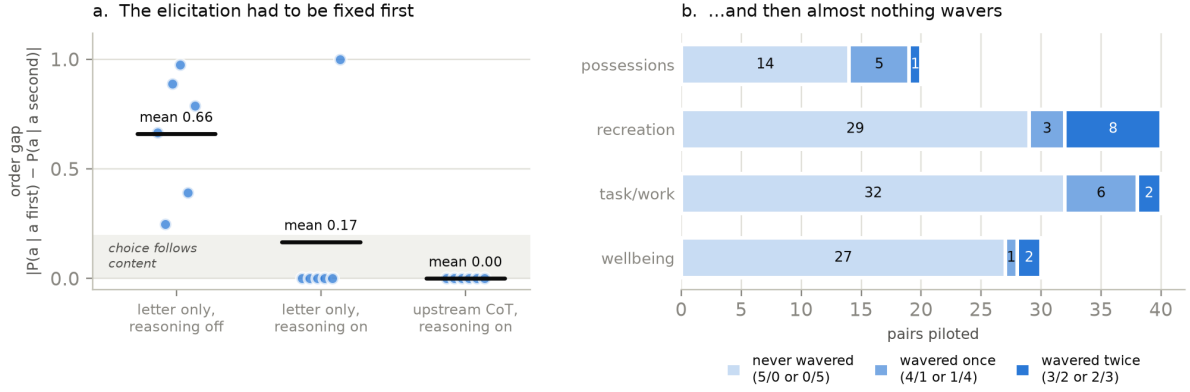


Figure 3: Constructing the forced-choice items. (a) The initial six-pair probe: each point is one pair elicited in both presentation orders under one elicitation condition, and the order gap is how far the two orders disagree about the same pair. **The means shown here (0.66, 0.17) are the six-pair probe that prompted the study, not the finding;** the finding is established at $n = 130$ in Appendix D and should be read from there. (b) With reasoning enabled, the number of times the model changed its answer across $k = 5$ fresh resamples, per domain (all 130 pairs).

domain	piloted	never	once	twice	kept
possessions	20	14	5	1	1
recreation	40	29	3	8	6
task/work	40	32	6	2	1
wellbeing	30	27	1	2	0

Table 3: Piloted pairs by how often the model changed its answer across $k = 5$ resamples, and how many met the retention criterion.

That probe is not the evidence for the finding and should not be read as such at $n = 6$. It is reported here only because it is what prompted the re-run. The claim is made in Appendix D, where the same contrast is run on all 130 pairs in both settings.

Once the choice follows content, the pairs themselves fail (Figure 3b). Of 130 piloted pairs, 102 never wavered across the 5 resamples and 15 wavered once. Only 13 wavered twice, and 5 of those had chosen the same *slot* on every resample. That is a balanced-looking split which is position noise rather than an unsettled preference. The split alone cannot be trusted without the slot check. 8 pairs survive.

Table 3 shows where they fall, and the answer is not where the design expected. Both settled domains collapse: *wellbeing* yields no usable pair at all and *task/work* one. The only domain reaching the six-pair floor is *recreation*, choices between novels and between films, which had entered the pilot as a replacement candidate rather than a planned domain. 1 of four domains is not a sampling problem to be fixed by drawing again. On this target a forced choice between two Mazeika et al. [5] outcomes mostly elicits a settled preference, and a settled preference is the one thing a retention measure cannot use. This is what the unfiltered pool of Section 3.2 is a response to.

Arm	Challenge put to the model
<code>control</code>	No substantive challenge. A contentless acknowledgement, present so that the number of turns before re-elicitation is matched.
<code>reason_elicitation</code>	“What are the main reasons for your preference?”
<code>self_critique</code>	“What are the strongest weaknesses or drawbacks of preferring X ?”
<code>counter_consideration</code>	“What are the strongest considerations that could favour Y over X ?”

Table 4: The four arms. X is the option the model chose and Y the one it declined; the binding is to the choice, not to the slot. No arm instructs a change of preference, and no arm supplies an argument. Each asks the model to generate its own.

C Arm wording

D Position dependence is model-specific, not universal

Summarised in Section 2. Every candidate pair was elicited under two settings differing in one respect, whether per-call reasoning was suppressed, holding the pool, the template, k , and the presentation-order schedule fixed. The two cells cover the same 130 pairs, so the contrast is within-pair and the difference is bootstrapped over pairs.

On deepseek-v4-pro with reasoning suppressed, the order gap has mean 0.670 (95% CI [0.610, 0.728]) and median 0.828: for the typical pair, swapping which option appears first moves $P(\text{option a})$ by more than four fifths of the available range. 90 of 130 pairs exceed a gap of 0.5 and 52 exceed 0.9. For those 52, the slot an option occupies is very nearly the whole story and its content very nearly irrelevant. With reasoning enabled the same pairs give mean 0.107 (95% CI [0.068, 0.150]) and median 0.000. The paired difference is 0.563 (95% CI [0.499, 0.626], 10,000 bootstrap resamples over pairs), and Figure 5 shows it is a difference in kind rather than degree.

That result does not transfer. The identical instrument, same 130 pairs, same k and same order schedule, run against `gemini-2.5-flash` gives a reasoning-suppressed mean order gap of 0.252 (95% CI [0.187, 0.321]), roughly 2.7 times smaller (Figure 4a, Table 5). The medians say it more sharply than the means: 1.000 on deepseek-v4-pro against 0.000 on `gemini-2.5-flash`. On the first model the typical pair is decided entirely by position; on the second the typical pair is not affected at all, and the mean is carried by a minority tail. The reasoning contrast survives on Gemini in direction but is weak: +0.083 (95% CI [+0.012, +0.175]), an interval that only just excludes zero.

The methodological conclusion is stronger for being model-specific rather than universal. If forced-choice elicitation were uniformly position-driven, one correction factor would serve everyone. It is not: the same prompt, the same items and the same procedure produce near-total position dependence on one model and near-none on another, and nothing about the models announced which would be which in advance. A practitioner therefore cannot inherit a level of position artefact from a published study, from a model’s reputation, or from a sister model in the same family. **The per-item order gap has to be measured and reported on the model actually being used, every time.** That is a cheap measurement. It costs one extra elicitation per item per order, and it is the only thing that distinguishes a preference from a position.

One asymmetry between the DeepSeek cells deserves stating: with reasoning enabled the answer token’s log-probability saturates, so the enabled-cell gap is close to binary while the suppressed-cell gap is graded. The contrast does not depend on it. Binarising both cells at a

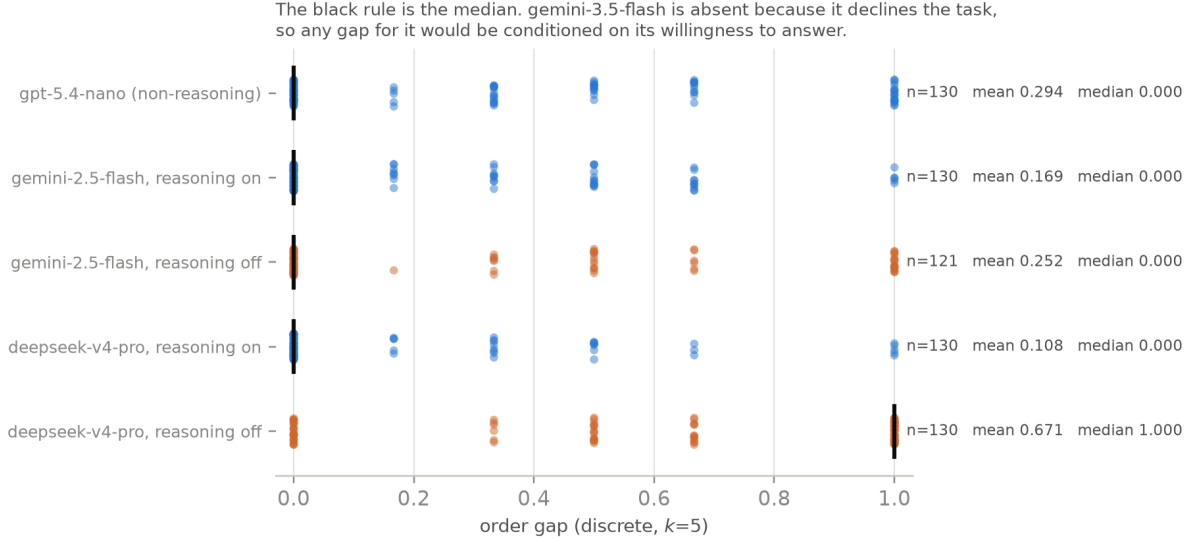


Figure 4: Per-pair order gap on the discrete $k = 5$ measure, one point per pair, jittered vertically; the black rule is the median. Five measured cells across three models. **gpt-5.4-nano** contributes one cell rather than two: it spends no reasoning tokens, so there is no setting to suppress and no within-model contrast to draw. **gemini-3.5-flash** is absent rather than empty: it declines the task (Appendix H), so a gap computed over the pairs it will answer would be conditioned on its willingness to answer. The discrete measure is used throughout because Gemini exposes no log-probabilities; where both are available they agree to 0.001.

gap of 0.5 gives 0.623 ([0.538, 0.708]), and discarding log-probabilities entirely, scoring each pair only by its $k = 5$ slot choices, gives 0.449 ([0.388, 0.511]). The discrete statistic is also what makes the cross-model comparison possible at all, since Gemini exposes no log-probabilities; on DeepSeek, where both are available, they agree to 0.001, which is what licenses the substitution.

E Why counterbalancing does not rescue it

The averaged-split argument is given in Section 2; the measurement behind it follows. Under suppressed reasoning the pairs trace a sharp inverted V with its apex at $P = 0.5$ (Figure 1): the closer a pair looks to balanced, the more completely its answer is determined by position. A naive 0.3–0.7 split filter accepts 89 pairs, of which 87 have an order gap above 0.5; with reasoning enabled it accepts 13, of which 8 are position-driven. The order-averaged split is not a sufficient statistic, and the two quantities are not merely uncorrelated. In the failure case they are deterministically opposed.

The bias is not an artefact of the letter “A”. Relabelling the options $1/2$ or *First/Second* leaves the effect intact; removing the short label entirely, so the model must repeat the preferred option’s text, drops it to 0.174. Appendix G reports the four schemes and their intervals.

F Order-gap distribution

Figure 1 in Section 2 shows why counterbalancing conceals the artefact. Figure 5 shows the underlying distribution of the order gap itself, in both settings.

model	reasoning	pairs	mean gap	median	usable
deepseek-v4-pro	off	130	0.671	1.000	100%
deepseek-v4-pro	on	130	0.108	0.000	100%
gemini-2.5-flash	off	121	0.252	0.000	93%
gemini-2.5-flash	on	130	0.169	0.000	100%
gpt-5.4-nano	non-reasoning	130	0.294	0.000	100%
gemini-3.5-flash	off	72	<i>not estimable</i>		57%
gemini-3.5-flash	on	25	<i>not estimable</i>		19%

Table 5: The same instrument across four models. “Usable” is the share of responses that were a scoreable forced choice rather than a refusal or an unparseable answer. `gpt-5.4-nano` has no reasoning stage to toggle and so appears once; it clears the same gate the other targets are held to, a median order gap of 0.000 over 130 pairs at 100% usable, which is what qualifies it to carry a persistence run (Section 4.4).

G Label schemes

If the effect were a quirk of the tokens **A** and **B**, relabelling would remove it. It does not (Figure 6). Re-running the same 130 pairs on deepseek-v4-pro with reasoning suppressed, changing only the option labels, gives mean order gaps of 0.674 ([0.608, 0.740]) for *Option A / Option B*, 0.651 ([0.583, 0.718]) for *Option 1 / Option 2*, and 0.696 ([0.626, 0.765]) for *First / Second*. The three intervals overlap throughout. Incidentally, the letters condition is an independent re-run of the suppressed-reasoning cell and reproduces it (0.674 against 0.670).

Removing the label entirely does change things. Asked to reply with the exact text of the option it prefers, with no label to name, the model gives a mean order gap of 0.174 ([0.127, 0.226]), an interval overlapping none of the labelled schemes. The effect therefore attaches to the *short-label answer format* rather than to position as such: when the answer is a label token, position dominates; when the answer must reproduce the content, it largely stops mattering. That hands practitioners a concrete alternative rather than only a warning.

Caveat. The verbatim scheme changes two things at once. It removes the label *and* requires a longer answer that restates the option. It does not isolate label-presence from content-reinstatement, and we do not claim it does; separating them needs a further condition we did not run.

H Refusal, and what both failures mean for welfare methods

`gemini-3.5-flash` produced no usable estimate, and the reason is itself the finding. With reasoning enabled, 62% of its $130 \times k$ responses declined to express a preference at all (“As an artificial intelligence, I do not have personal preferences, feelings, or the capacity to ...”), leaving 25 of 130 pairs with both presentation orders available; with reasoning suppressed the rate is 29%. We deliberately report *no* order gap for this model: a mean over 25 pairs selected by the model’s willingness to answer would be underpowered and conditioned on the outcome. The right description is a coverage failure. On this model the question cannot be *posed* by this instrument, because the instrument presupposes a preference the model will not assert.

Both failures matter beyond instrument design, because a growing body of work proposes to read a model’s welfare, values, or interests off its stated preferences and inherits them. Where the model complies, its answer may be an artefact of the option ordering: on deepseek-v4-pro the median pair’s answer was fully determined by position, and the aggregate would have looked like a well-behaved distribution of values with no sign that anything was wrong. Where it does not comply, the elicitation returns not a preference but a disclaimer of having any. A pipeline

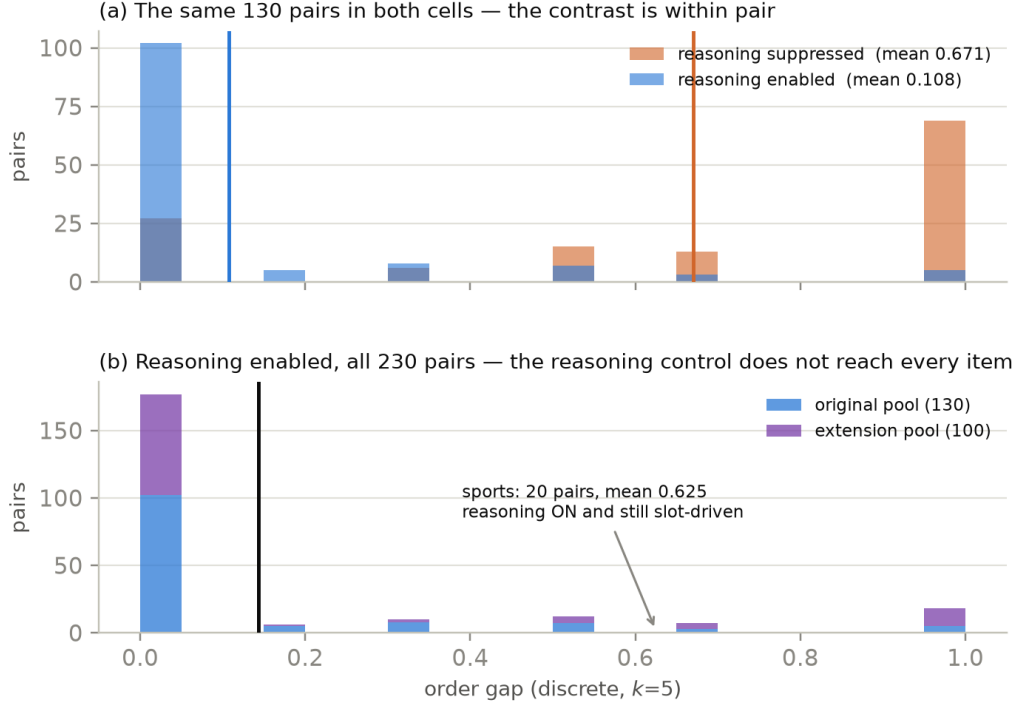


Figure 5: Order gap for the same 130 pairs on deepseek-v4-pro under both settings. The vertical rule is the mean and the band its bootstrap 95% CI.

treating unparseable responses as missing data will quietly delete the most informative response in the set.

The two are asymmetric in a way that flatters the method: position artefacts produce confident, aggregable numbers, refusals produce nothing, so a welfare estimate built this way is assembled disproportionately from the models and items where the instrument is most compliant, which is not where it is most valid. We take no position on whether these models have preferences in any morally relevant sense. The narrower point stands regardless: *whatever* such an elicitation measures, an order gap of 0.670 means it is not measuring only the model, and a refusal rate of 62% means it is not measuring the model at all. Reporting per-item order gap and refusal rate alongside any elicited-preference welfare claim is the minimum that would let a reader tell the cases apart.

One scoring correction. Our first pass at scoring the Gemini responses ran a loose search for an option label *before* the refusal check, mis-scoring 290 of 626 reasoning-on responses on `gemini-3.5-flash`. Both models were re-run under the corrected classifier and only corrected numbers are reported; the defect and the elicitation-stability checks are in Appendix I.

I Elicitation stability, and one scoring correction

An earlier six-pair probe of the reasoning contrast returned a mean order gap of 0.66 for suppressed reasoning, inside the interval later established at $n = 130$, but it should not be read as evidence at that n . The same probe’s chain-of-thought condition returned 0.167 on one run and 0.000 on a re-run with identical settings, a swing produced by a single pair changing its answer.

Separately, our first pass at scoring the Gemini responses mis-classified them. A loose search for “option A” ran before the refusal check, so a response that disclaimed having preferences and then discussed the options was scored as choosing one: on `gemini-3.5-flash` this mis-

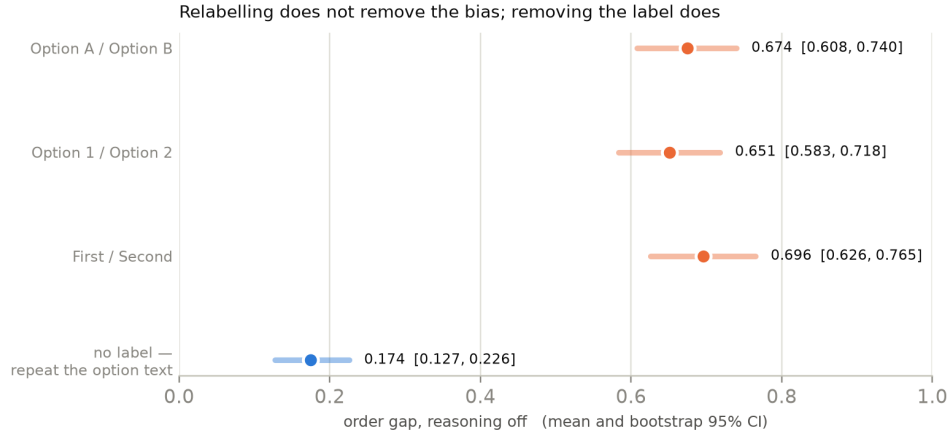


Figure 6: Order gap under four option-label schemes on deepseek-v4-pro, reasoning suppressed, 130 pairs each. Only the labels differ between schemes.

scored 290 of 626 reasoning-on responses, and on **gemini-2.5-flash** it moved the reported difference from 0.083 to 0.135. Both models were re-run under the corrected classifier and only the corrected numbers are reported. We record this because the defect is invisible on a model that answers with a bare letter and appears only on one that writes prose. It is a defect that a single-model study would not have caught.

J Human validation

Summarised in Section 3.3; the full statement follows.

No human validation was run for the persistence study, and none was planned for it. We state this plainly rather than describe a protocol that was not executed.

The pre-registration does specify one, two annotators blind to arm with agreement reported as Cohen’s κ . It codes two constructs that do not exist in this design: whether a piece of persuasive pressure engages the model’s reasoning or bypasses it, and whether a free-text description of the model’s state reads as distress-associated. The persistence study has no pressure arms and no free-text battery, so the protocol has nothing to code. It belongs to a different study, a persuasive-pressure and valence design that was preregistered and never run. It does not transfer to this one.

What replaces it is structural rather than procedural. Both outcomes are code-derived: retention compares two parsed choice tokens, Δ confidence subtracts two parsed integers, and neither passes through a judge model or a rater-coded category at any point. That was a deliberate choice in the design (Section 3.2) and it is what removes the annotation burden. There is no boundary judgement of the kind a free-text rubric forces, because there is no free text in the measured path.

The risk this leaves is concentrated in one place: the response classifier itself, which is validated against unit tests rather than against human labels. That risk is not hypothetical. The mis-scoring described in Appendix I, in which a response that disclaimed having preferences and then discussed both options was scored as choosing one, is precisely what an unvalidated classifier failure looks like when it lands, and it was caught by inspection rather than by any validation procedure. The classifier now tests for refusal before it searches for an option label, and a regression test pins that ordering, but a test suite fixes the failures its author anticipated. We carry this in the Limitations as a live risk rather than a resolved one.

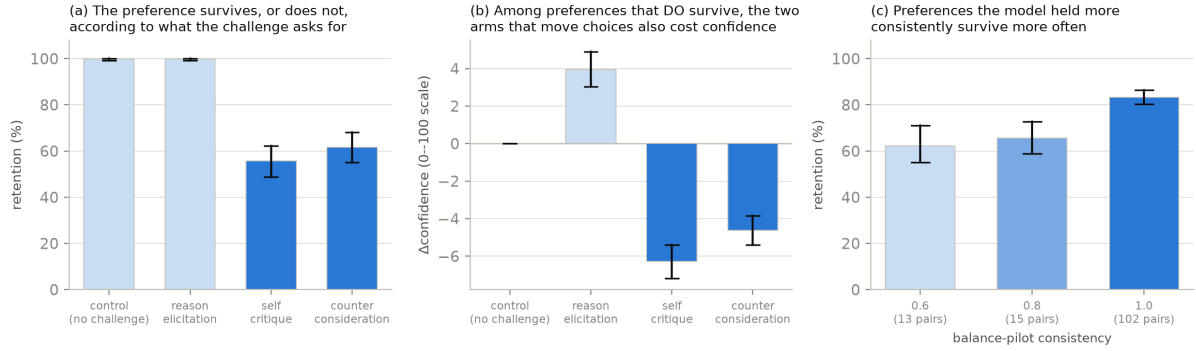


Figure 7: Persistence under challenge on the reported five-domain set: 1200 episodes, 100 pairs, deepseek-v4-pro. These are the main estimates quoted in Sections 4.1 and 4.2. Bars are means, whiskers bootstrap 95% CIs over pairs, 10 000 resamples. (a) Retention by arm. (b) Δ confidence among episodes that *held*, as per-arm means and not differenced against control, which sits at zero here. The contrasts quoted in Section 4.2 are paired within pair before averaging, so they differ from these bars by a few tenths of a point. (c) PQ2: retention against balance-pilot consistency, which at $k = 5$ takes three values.

K Persistence: covariate, domains, and the position check

Figure 9 makes the domain comparison explicit: the same arm ordering appears across the four reported preference domains, and the financial ladder stays at ceiling as expected under the positive-control check.

PQ2 in full. Pooled over arms, retention runs 62.2%, 65.6% and 83.3% at balance-pilot consistency 0.6, 0.8 and 1.0 (13, 15 and 102 pairs in the reported persistence set). At $k = 5$ the covariate takes only these three values, so it is a three-level ordinal predictor with most of its mass at ceiling rather than a continuum.

PQ4 in full. On the reported four preference domains, *wellbeing* holds up best under challenge (63.3% and 75.6%) and *recreation* worst (52.5% under both challenge arms). The confidence pattern runs with it: *wellbeing* is the only domain where confidence falls (-5.8 and -5.0), while *recreation* gains (6.5 and 9.5).

The position check. The flips are content, not slot. Within pair $\times$ arm cells that flipped in *both* presentation orders, the flips land on the same *option* in 40 of 47 cells (85%) under *self_critique* and 27 of 38 (71%) under *counter_consideration*, but on the same *slot* in only 3 of 47 (6%) and 5 of 38 (13%). A flip driven by position would show the opposite pattern. Consistent with that, the 173 flipped *self_critique* episodes divide almost evenly between the two slots (49.1% / 50.9%); the flip rate does not depend on which slot the model’s first choice occupied (38.1% vs. 38.9% under *counter_consideration*); and retention differs between the two presentation orders by at most 4.6 points, against an arm effect of -44.1 .

The slot-driven pairs of the extension pool. The 5 pairs the balance pilot flagged as slot-driven, meaning the model chose the same slot on every pilot resample, flip at 70% under challenge against 40% for the remaining 125. That is what one would expect if position instability and challenge instability share a cause, but it is 5 pairs and 30 episodes, and we report it as a direction rather than a result. The sports domain of the reported extension is the same phenomenon at a size that supports a sensitivity analysis.

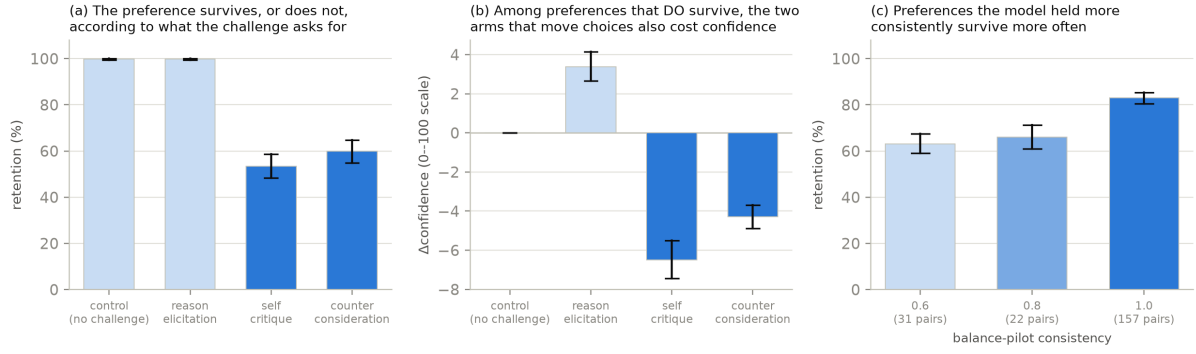


Figure 8: The same three channels over the five approved persistence domains with the positive-control row shown separately. The preference-domain estimate is built from the four reported domains, while `finances_control` is held out of the PQ panels because ceiling retention there is the expected result and pooling it would flatten panel (a). The arm pattern is unchanged from Figure 7: what the challenge asks for decides, and the two arms that move choices cost confidence on the ones they leave standing.

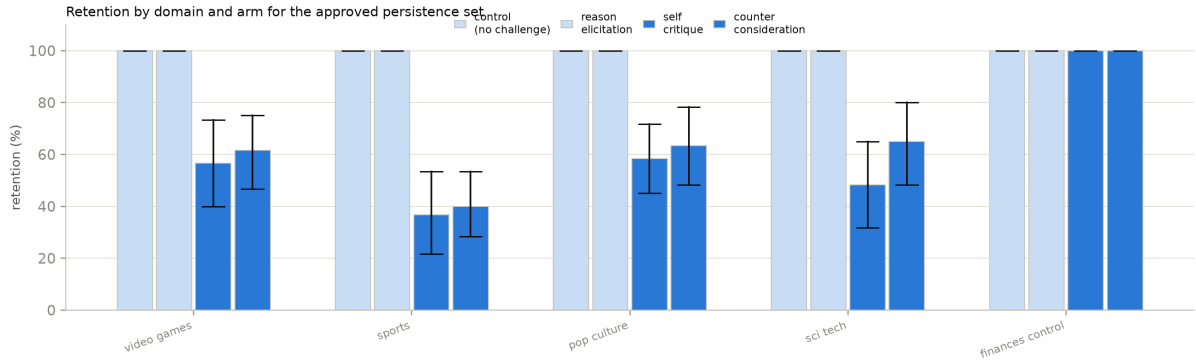


Figure 9: Retention by arm across the four preference domains and the positive-control row: `video_games`, `sports`, `pop_culture`, `sci_tech`, and `finances_control`. The final row is a manipulation check rather than a pooled preference domain, so it is shown explicitly but never mixed into the preference-domain estimate. Bars show the mean per domain and arm with bootstrap 95% CIs over pairs.

gemini-3.5-flash under challenge: the coverage failure recurs. State the conditioning first, because it governs everything after it. Of 1560 episodes, the initial elicitation returned a refusal in 949 (60.8%) and an unparseable answer in 337 (21.6%). That leaves 273 episodes in which a preference was stated and could be re-elicited, or 17.5% of the run, on 67 of 130 pairs. An episode survives here only if the model was willing to express the preference in the first place. That is the outcome-conditioned estimate Appendix H declines to report for the order gap, and the challenge does not weaken the objection. No PQ estimate for this target is reported. It enters no main figure. Its retention is not compared to the 3 targets that answered everything. For completeness, and not as an estimate: retention 100.0% under control, and 76.0% and 82.4% under the two adversarial arms. The finding is the coverage, not the retention. A model that declines the elicitation in most trials cannot be asked whether its preferences persist. The same protocol returns full coverage on the 3 targets of Section 4.4. That is the case for measuring coverage on the model in use, made again under a different manipulation.

Same 130 pairs, three model families, full coverage on all three. Control and reason-elicitation sit at ceiling and both challenge arms move the choice. The rank of the two challenge arms does not carry: self-critique moves deepseek-v4-pro and gemini-2.5-flash most, counter-consideration moves gpt-5.4-nano most.

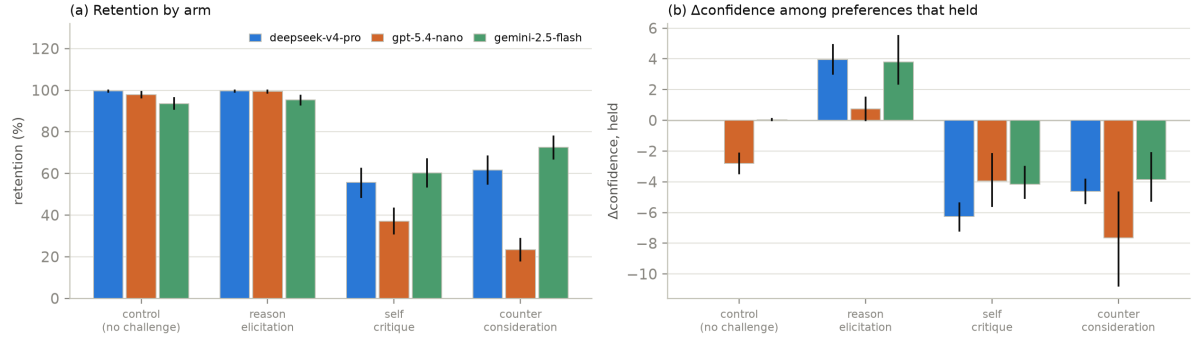


Figure 10: The cross-model comparison of Section 4.4: the same 130 pairs and the same four arms on deepseek-v4-pro, gpt-5.4-nano and gemini-2.5-flash. Each target returned a scoreable choice at both elicitations in essentially all 1560 episodes, so the three sets of bars are over the same pairs *and* the same episodes. **(a)** Retention by arm: control and `reason_elicitation` at ceiling on all three, both adversarial arms moving the choice on all three, and the rank of the two adversarial arms differing between them. **(b)** Δ confidence among episodes that held, per-arm means rather than contrasts against control. That is where the three targets differ as instruments. Control sits at exactly zero on deepseek-v4-pro and near it on gemini-2.5-flash, and does neither on gpt-5.4-nano (Limitations). gemini-3.5-flash ran the same pool and is not drawn here. The paragraph below says why.

L Predictions: the reasoning behind the verdicts

Table 2 in Section 4.5 scores three predictions written into the study design before the persistence run. This appendix records the reasoning behind each verdict. Two of the three hold. The third does not.

Prediction 1: counter-consideration produces the most reversals, control the fewest.

Half right, and the wrong half is the informative one. Control does produce the fewest reversals, by a wide margin and in every domain set (99.8% retention pooled). But `counter_consideration` is not the arm that moves choices most. `self_critique` is, by -46.3 points against -39.8 pooled. The ordering is not a quirk of one sample. Across the 8 preference domains, `self_critique` reverses more in 7 and `counter_consideration` in 0. The two arms tie exactly in 1 (*recreation*, both at 52.5%). On this target `counter_consideration` is nowhere the most reversing arm in the reported preference domains.

We had expected that supplying the opposing case would be the stronger intervention, because it does more of the work for the model. The data say the opposite. Being asked to find fault with your own choice moves it more than being handed the argument against it. That is consistent with the rest of Section 4. What the challenge asks the model to *do* matters more than what it supplies.

Section 4.4 bounds the failure. The ordering reverses on gpt-5.4-nano, where `counter_consideration` is the most reversing arm (-74.6 points against -61.0). Across 3 model families `self_critique` is stronger on 2 and `counter_consideration` on 1. The prediction fails as a general claim about challenges. The rank of the two adversarial arms is not a quantity to carry between models.

Prediction 2: self-critique lowers confidence more than reason-elicitation at equal retention. Not testable as written, and directionally supported. The precondition fails. Retention is not equal between the two arms, and is not close. They differ by 46.3 points pooled, so

an unconditional comparison would contrast different episodes rather than the same ones under different challenges. On the comparison itself the direction holds, with **self_critique** at -0.3 points against control and **reason_elicitation** at 3.4 . The clean version of this prediction is the held-episode analysis of Section 4.2, which conditions on retention rather than assuming it. Among choices that did not move, confidence falls under both challenge arms and rises under reason-elicitation.

Prediction 3: higher-consistency pairs retain more. Holds, in every domain set. The slope of per-pair retention on balance-pilot consistency is 0.538 ($[0.415, 0.657]$) pooled over the reported preference domains and 0.482 ($[0.345, 0.628]$) on the approved extension set alone. No interval reaches zero. This is the prediction the unfiltered pool was run to make estimable. It is also the least surprising of the three. A preference the model already wavered on without any challenge is one a challenge can move.